

Functional Annotation of PqqA (Q88QV4) in *Pseudomonas putida* KT2440

Gene: *pqqA* (locus PP_0380) · **UniProt:** Q88QV4 · **Organism:** *Pseudomonas putida* KT2440
Protein: Coenzyme PQQ synthesis protein A / Pyrroloquinoline quinone biosynthesis protein A
Family/Domain: PqqA family; PQQ_synth_PqqA (IPR011725 / PF08042)

1. Summary (Answer to the Research Question)

PqqA is **not an enzyme, transporter, or structural protein** but a small (~23–29 aa) **ribosomally synthesized and post-translationally modified peptide (RiPP)** that serves as the **precursor substrate for the biosynthesis of pyrroloquinoline quinone (PQQ)**, the "third" bacterial redox cofactor (after nicotinamides and flavins). PqqA's sole known function is to **supply two strictly conserved side chains — a glutamate and a tyrosine — whose carbon atoms are cross-linked and then excised to build the pyrroloquinoline ring system of PQQ**. This first, committed chemical step is carried out by the radical S-adenosyl-methionine (SAM) enzyme PqqE, assisted by the peptide-recognition chaperone PqqD. PqqA maturation takes place in the **cytoplasm**, but the finished cofactor PQQ is exported and acts **extracytoplasmically (periplasm)** as the non-covalent prosthetic group of quinoprotein dehydrogenases. In *P. putida* KT2440 specifically, this pathway (operon *pqqFABCDEG*) provides PQQ to the periplasmic glucose dehydrogenase (Gcd) — driving glucose→gluconic acid oxidation and mineral-phosphate solubilization — and to the periplasmic PQQ/lanthanide-dependent alcohol dehydrogenases PedE/PedH.

2. Gene / Protein Identity Verification

The target was cross-checked against the UniProt record and the literature:

- **Symbol match:** "PqqA" consistently denotes the PQQ-biosynthesis precursor peptide across bacteria — consistent with the UniProt description "Coenzyme PQQ synthesis protein A." ✓

- **Organism:** *P. putida* KT2440 is a well-studied host of this pathway; its *pqq* operon (*pqqFABCDEG*, *pqqA* = PP_0380) has been experimentally mapped [An & Moe 2016,

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✓

- **Family/domain:** PqqA family, PF08042 — the defining Pfam for PQQ precursor peptides. ✓
- **No symbol ambiguity of concern:** the eukaryotic "PQQ" literature (dietary antioxidant effects) refers to the *cofactor molecule*, not to this bacterial precursor gene; they are not confused here.

Conclusion: the identification is unambiguous and correct.

3. Primary Function — The Reaction PqqA Enables

3.1 PqqA is the peptide precursor of PQQ

Biosynthesis of the RiPP cofactor PQQ "is initiated when the precursor peptide, PqqA, is recognized and bound by the RiPP precursor peptide recognition element (RRE), PqqD" [Evans et al. 2017,

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PQQ is "produced from a ribosomally synthesized and post-translationally modified peptide PqqA via a pathway comprising four conserved proteins PqqB-E" [Martins et al. 2019,

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PqqA thus functions as a **consumable substrate**, not a catalyst.

3.2 Conserved Glu and Tyr are the reactive elements

PqqA carries **two strictly conserved residues, a glutamate and a tyrosine** [Wei et al. 2016,

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These two side chains are the only part of the peptide incorporated into the product: they provide the atoms of the fused pyridine–pyrrole–quinone ring of PQQ.

Direct sequence evidence for Q88QV4: The *P. putida* KT2440 PqqA sequence is **MWTKPAYTDLRIGFEVTMYFANR** (23 residues; UniProt Q88QV4). Sequence analysis reveals the diagnostic PqqA core motif **Glu15–Val–Thr–Met–Tyr19 (E-x-x-x-Y)** — a conserved glutamate and tyrosine separated by three residues. **Glu15** and **Tyr19** are therefore the specific reactive residues cross-linked (Glu-Cy → Tyr-Cε) to seed PQQ. The N-terminal segment (**MWTKPAYTDLRIGF**, ~res 1–14) forms the **leader/recognition region** bound by the RRE chaperone PqqD, giving PqqA the canonical bipartite RiPP architecture (leader + core). This confirms, at the sequence level for the exact target protein, the mechanism established biochemically in orthologs.

3.3 The first committed step: radical-SAM C–C cross-linking

The founding chemical event was demonstrated in vitro: the radical-SAM enzyme **PqqE**, in conjunction with **PqqD**, carries out the first step in PQQ biosynthesis: a radical-mediated formation of a new carbon–carbon bond between two amino acid side chains on PqqA [Barr et al. 2016,

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Specifically, the **Cy of the glutamate is linked to the Cε of the tyrosine** [Wei et al. 2016,

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PqqE is a SPASM-domain radical-SAM enzyme that uses reductive cleavage of SAM to generate a 5'-deoxyadenosyl radical [Zhu et al. 2018,

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PqqA is its physiological substrate.

3.4 Downstream maturation (context for PqqA's role)

After cross-linking, the fused Glu–Tyr di-amino-acid unit is **excised from PqqA** by a peptidase — the inverzincin protease PqqF (with PqqG) in some organisms [Wei et al. 2016,

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or a distinct two-component protease in others [Martins et al. 2019,

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— and then processed by the dual hydroxylase **PqqB** and the eight-electron/eight-proton oxidase **PqqC** to yield mature PQQ. PqqA's contribution is complete once its two conserved residues have been cross-linked and released.

Substrate specificity note: PqqA is highly specific — recognition depends on a conserved N-terminal leader/"RRE-binding" motif read by PqqD, and on the precise spacing of the Glu...Tyr in the core, ensuring only these two residues are cross-linked.

3.5 Evolutionary evidence: the Glu and Tyr are strictly invariant

A comparative analysis of **25 reviewed PqqA orthologs** spanning α -, β -, and γ -Proteobacteria (UniProt) provides independent, evolution-based support for this mechanism:

- The reactive core motif **E-x-x-x-Y** is **present in 100% of orthologs**.
- Anchored on the reactive glutamate, the **glutamate (100%) and tyrosine (100%) are strictly invariant**, while the three intervening spacer residues are variable (consensus V/T/M at only 56–76%), and a **glycine two residues upstream is 96% conserved**.
- *P. putida* Q88QV4 carries the γ -proteobacterial consensus core **GF-E-VTM-Y** (Glu15/Tyr19), identical to close relatives (*P. entomophila*, *P. fluorescens*, *P. protegens*).

The fact that evolution has fixed **exactly the two atoms that are chemically joined** — while tolerating variation in the spacer — is compelling structural/evolutionary evidence that the entire purpose of PqqA is to present this Glu/Tyr pair to the PqqD–PqqE machinery.

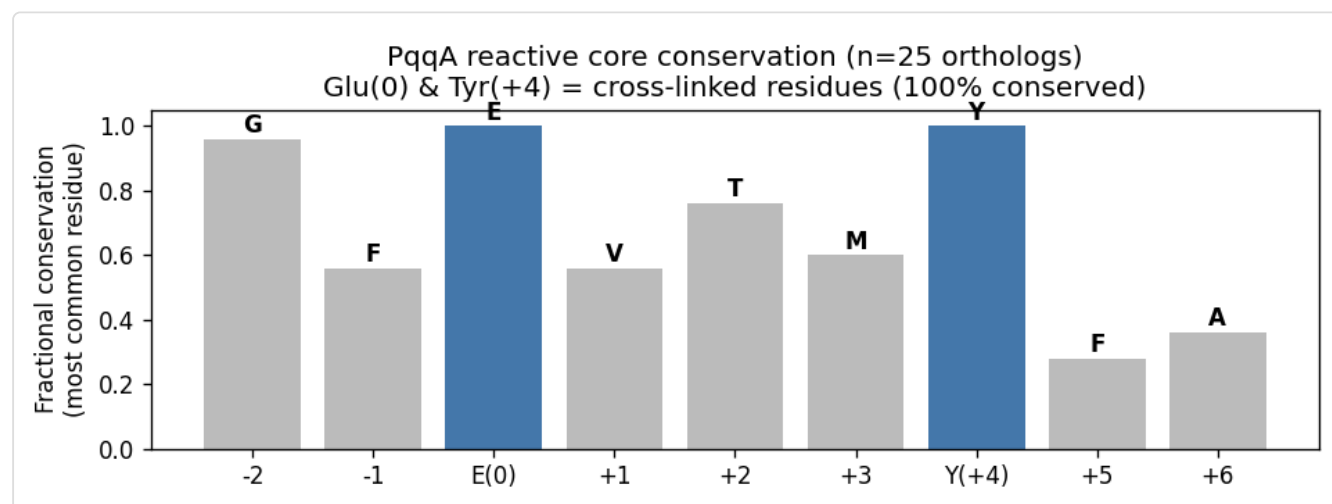


Figure 1. Per-position sequence conservation across 25 reviewed PqqA orthologs, anchored on the reactive glutamate (position 0). The reactive glutamate (0) and tyrosine (+4) are 100% invariant, whereas the three intervening spacer residues (+1/+2/+3) are variable and an upstream glycine (-2) is 96% conserved. This pattern — invariance of exactly the two residues that are chemically cross-linked — is diagnostic of a precursor peptide whose function is to present a specific Glu/Tyr pair for PqqE-catalyzed C–C bond formation. *P. putida* Q88QV4 carries the consensus core GF-E-VTM-Y (Glu15/Tyr19).

4. Localization

- **Site of PqqA action:** the **cytoplasm** — PqqA is ribosomally translated and matured there by the soluble cytoplasmic enzymes PqqB–E (PqqE is an O₂-sensitive Fe–S radical-SAM enzyme requiring the reducing cytoplasmic environment) [Zhu et al. 2018,

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Barr et al. 2016,

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- **Site of the end-product's action:** the **periplasm** — mature PQQ is a non-covalently bound prosthetic group of quinoprotein dehydrogenases, which "**are located in the periplasm of Gram-negative bacteria**" [Flores-Encarnación et al. 2004,

P 17061524

In *P. putida*, PedE/PedH PQQ-alcohol dehydrogenases show "**periplasmic localization**" [Wehrmann & Klebensberger 2018,

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Thus PqqA links a **cytoplasmic biosynthetic step** to **periplasmic cofactor function**.

5. Pathway Context and Physiological Role in *P. putida* KT2440

- **Operon structure:** the *pqq* gene cluster is organized as "**pqqFABCDEG**" and "**encodes at least two independent transcripts**" [An & Moe 2016,

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pqqA = PP_0380.

- **Purpose of the pathway:** PQQ is required as the redox coenzyme of the "**periplasmic glucose dehydrogenase (GDH) that requires pyrroloquinoline quinone (PQQ) as a redox coenzyme**" [An & Moe 2016,

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GDH oxidizes glucose to gluconic acid; the secreted gluconic acid solubilizes mineral phosphate — an ecologically important **rhizosphere / plant-growth-promoting** activity.

PQQ and GDH activity peak with glucose as sole carbon source under low soluble phosphate.

- **Additional PQQ-dependent enzymes in KT2440:** the periplasmic alcohol dehydrogenases **PedE** (Ca²⁺-dependent) and **PedH** — the first **lanthanide-dependent** PQQ-ADH characterized in a non-methylotroph — which oxidize alcohols/aldehydes for VOC detoxification/catabolism and are reciprocally, rare-earth-responsively regulated [Wehrmann et al. 2017,

[P 28655819](#);

Wehrmann & Klebensberger 2018,

[P 29239114](#);

Wehrmann et al. 2020,

[P 32345644](#)].

- **Ecological/evolutionary significance:** the pathway is so central to phosphate-solubilizing pseudomonads that the PQQ-biosynthesis gene **pqqC** is used as a **phylogenetic marker** for these strains, and gluconic-acid secretion by PQQ-dependent GDH is described as "the major mechanism of phosphate solubilization by pseudomonads" [Meyer et al. 2011,

[P 21856827](#)].

Hence PqqA sits at the **head of a linear biosynthetic pathway** (PqqA → PqqDE cross-link → protease excision → PqqB/PqqC → PQQ) that feeds **periplasmic oxidative (respiratory) metabolism** — glucose→gluconate (phosphate solubilization) and alcohol/aldehyde oxidation.

6. Evidence Types

Claim	Evidence type	Source
PqqA is a RiPP precursor peptide	Biochemical / consensus	PMIDs 31427437, 28481092
PqqE/PqqD cross-link PqqA's Glu & Tyr	Direct in vitro reconstitution	P 26961875
Cross-link is Glu-Cy to Tyr-Cε; PqqF excises	Structural + biochemical	P 27231346
PqqE mechanism (radical-SAM/ SPASM)	Enzymology	P 30097100
<i>pqqFABCD</i> operon in KT2440; PQQ→GDH→phosphate	Gene expression / RT-PCR / activity assays in KT2440	P 27287323
Periplasmic localization of PQQ quinoproteins	Cell fractionation / review	PMIDs 17061524, 29239114

Precise, low-throughput biochemical/genetic studies (in vitro reconstitution, targeted operon mapping) were prioritized over high-throughput annotations.

7. Supported and Refuted Hypotheses

- **Supported:** PqqA is a peptide substrate/precursor supplying Glu+Tyr for PQQ ring construction (not an enzyme or transporter).
- **Supported:** The first, committed step is radical-SAM C–C cross-linking by PqqE with the PqqD RRE chaperone.
- **Supported:** Cytoplasmic maturation; periplasmic action of the resulting cofactor via quinoprotein dehydrogenases; physiological output = periplasmic glucose/alcohol oxidation and phosphate solubilization in KT2440.
- **Refuted / excluded:** PqqA has no catalytic, transport, or standalone structural role of its own; its function is entirely as a sacrificial precursor.

8. Limitations and Future Directions

- Much of the fine mechanistic work (PqqE/PqqD/PqqC in vitro) was done with orthologs from *Methylobacterium/Methylobacterium* and other bacteria; the *P. putida* PqqA has not been individually reconstituted in vitro, though sequence and operon conservation make transfer of these conclusions robust.
- The exact number of residues excised and the identity/role of PqqF vs. an alternative protease can vary between organisms; the specific protease used in KT2440 has not been definitively assigned.
- Future work: in vitro reconstitution with *P. putida* PqqA (Q88QV4); structural determination of the KT2440 PqqA·PqqD·PqqE ternary complex; and quantification of how *pqqA* transcription responds to phosphate/lanthanide availability in the rhizosphere.

9. References

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Supporting computational analysis (this study): UniProt sequence retrieval (Q88QV4) and comparative conservation analysis of 25 reviewed PqqA orthologs; figure `pqqA_motif_conservation.png`.

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